

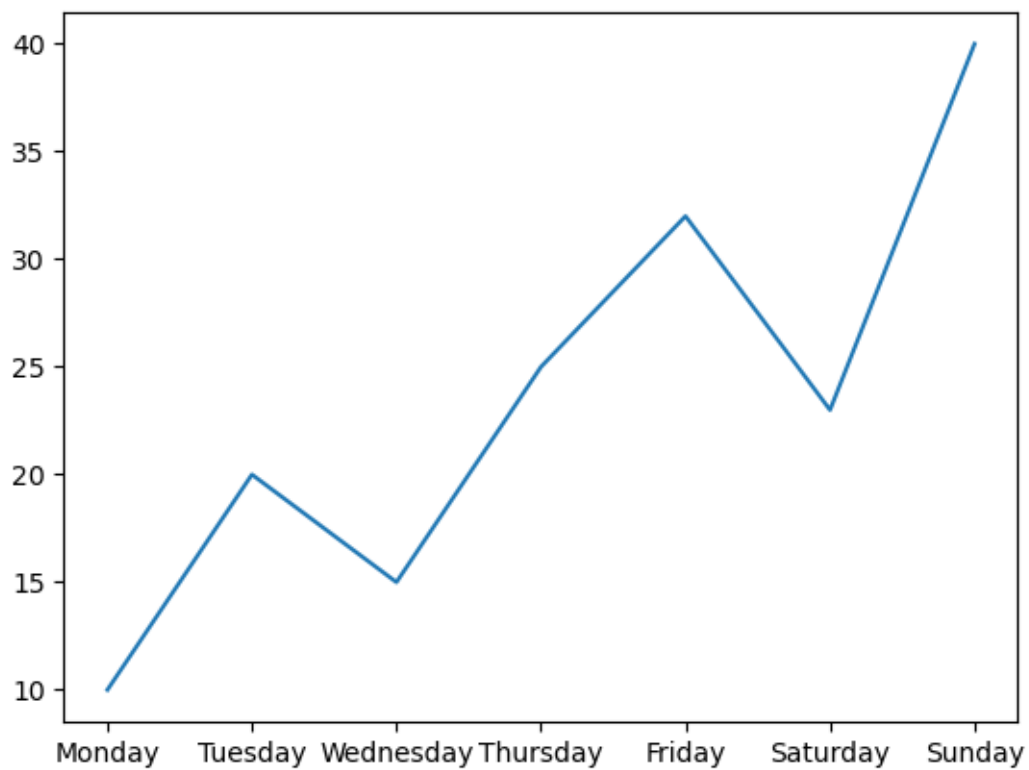
Matplot_Lib

September 30, 2025

```
[1]: import matplotlib.pyplot as plt

# x=[1,2,3,4]
x=['Monday', 'Tuesday', 'Wednesday', 'Thursday', 'Friday', 'Saturday', 'Sunday']
y=[10,20,15,25,32,23,40]

plt.plot(x,y)
plt.show()
```



```
[5]: import matplotlib.pyplot as plt

# x=[1,2,3,4]
```

```

x=['Monday','Tuesday','Wednesday','Thursday','Friday','Saturday','Sunday']
y=[10,20,15,20,32,23,40]

plt.plot(x,y)

plt.title("Bakery Products sell in a week!")
plt.xlabel('Days')
plt.ylabel('Cakes Sold')
plt.show()

```



```

[4]: import matplotlib.pyplot as plt

# x=[1,2,3,4]
x=['Monday','Tuesday','Wednesday','Thursday','Friday','Saturday','Sunday']
y=[10,20,15,20,32,23,40]

plt.plot(x,y,color='red',linestyle='-',linewidth=1,marker='o')

# plt.
    ↪ plot(x,y,color='red',linestyle='-',linewidth=1,marker='marker',label='label_name')

```

```
plt.title("Bakery Products sell in a week!")
plt.xlabel('Days')
plt.ylabel('Cakes Sold')
plt.show()
```



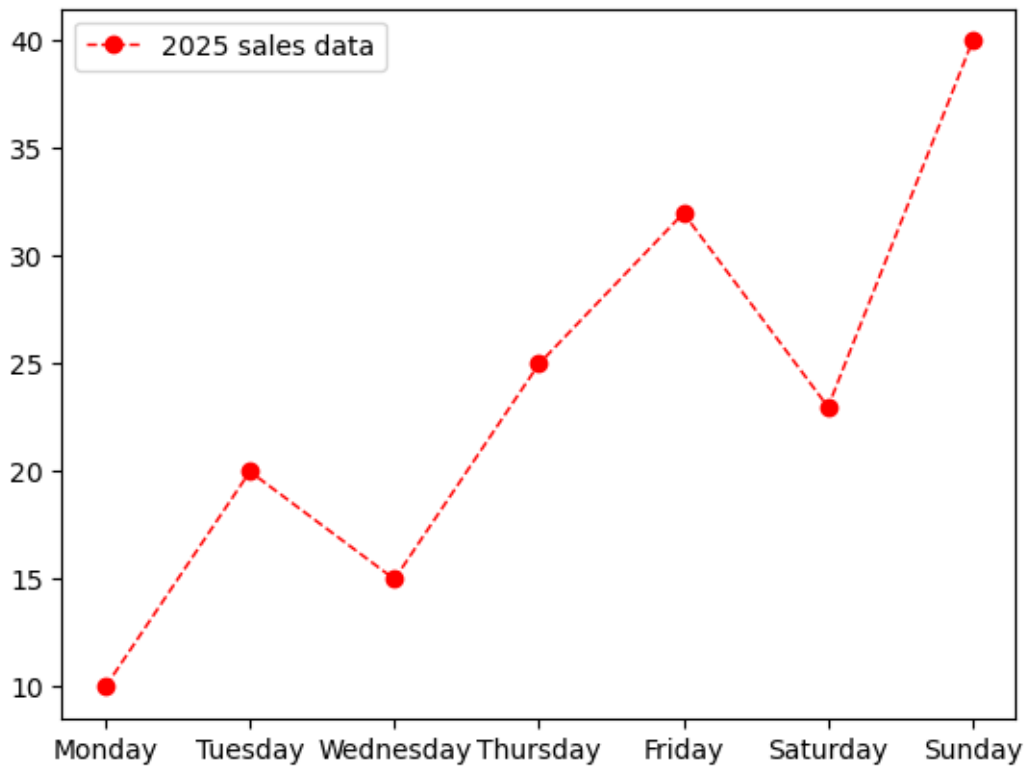
```
[8]: import matplotlib.pyplot as plt

# x=[1,2,3,4]
x=['Monday', 'Tuesday', 'Wednesday', 'Thursday', 'Friday', 'Saturday', 'Sunday']
y=[10,20,15,25,32,23,40]

plt.plot(x,y,color='red',linestyle='--',linewidth=1,marker='o',label='2025_
↳sales data')

# plt.
↳plot(x,y,color='red',linestyle='--',linewidth=1,marker='marker',label='label_name')
# plt.title("Bakery Products sell in a week!")
# plt.xlabel('Days')
# plt.ylabel('Cakes Sold')
```

```
plt.legend()
plt.show()
```

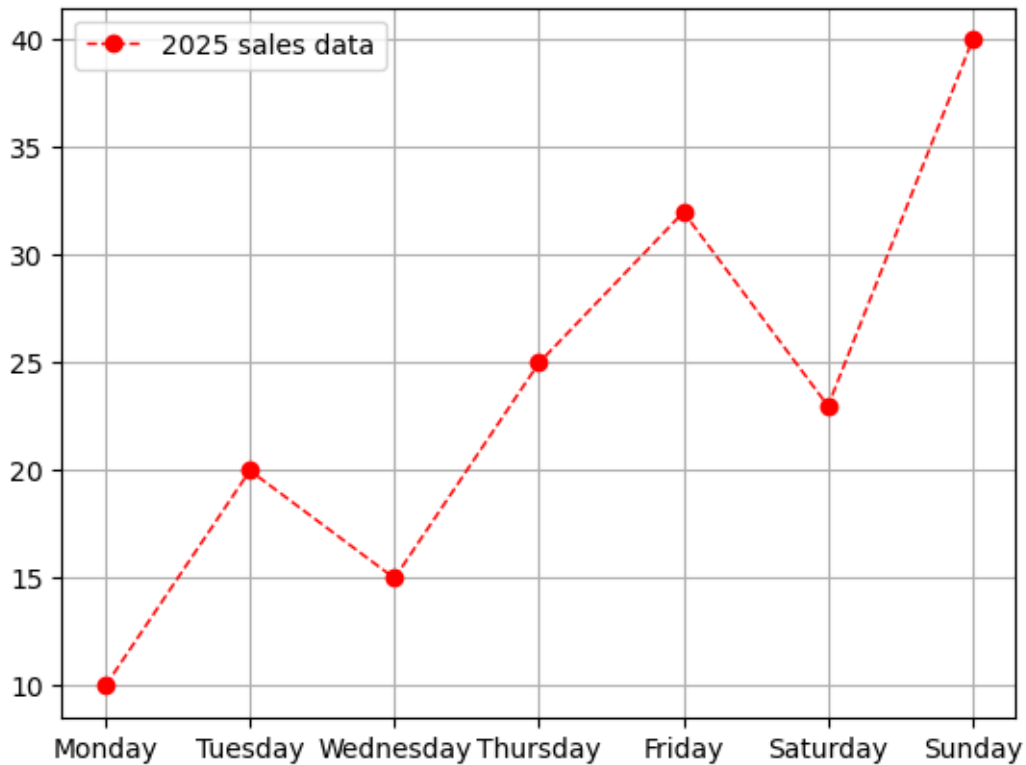


```
[9]: import matplotlib.pyplot as plt

# x=[1,2,3,4]
x=['Monday', 'Tuesday', 'Wednesday', 'Thursday', 'Friday', 'Saturday', 'Sunday']
y=[10,20,15,25,32,23,40]

plt.plot(x,y,color='red',linestyle='--',linewidth=1,marker='o',label='2025_
↳sales data')

# plt.
↳plot(x,y,color='red',linestyle='-',linewidth=1,marker='marker',label='label_name')
# plt.title("Bakery Products sell in a week!")
# plt.xlabel('Days')
# plt.ylabel('Cakes Sold')
plt.legend()
# plt.legend(loc='upper left/lower right',fontsize=12)
plt.grid()
plt.show()
```

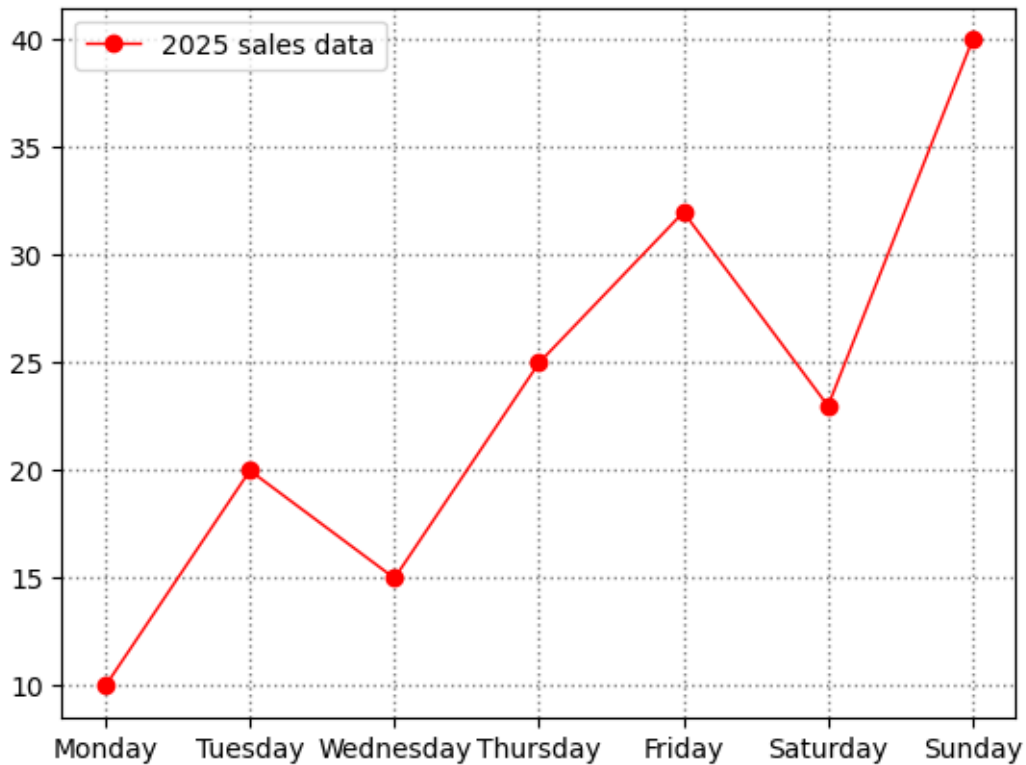


```
[11]: import matplotlib.pyplot as plt

# x=[1,2,3,4]
x=['Monday', 'Tuesday', 'Wednesday', 'Thursday', 'Friday', 'Saturday', 'Sunday']
y=[10,20,15,25,32,23,40]

plt.plot(x,y,color='red',linestyle='-',linewidth=1,marker='o',label='2025 sales_
↳data')

# plt.
↳plot(x,y,color='red',linestyle='-',linewidth=1,marker='marker',label='label_name')
# plt.title("Bakery Products sell in a week!")
# plt.xlabel('Days')
# plt.ylabel('Cakes Sold')
plt.legend()
# plt.legend(loc='upper left/lower right',fontsize=12)
# plt.grid()
plt.grid(color='gray',linestyle=':',linewidth=1)
plt.show()
```

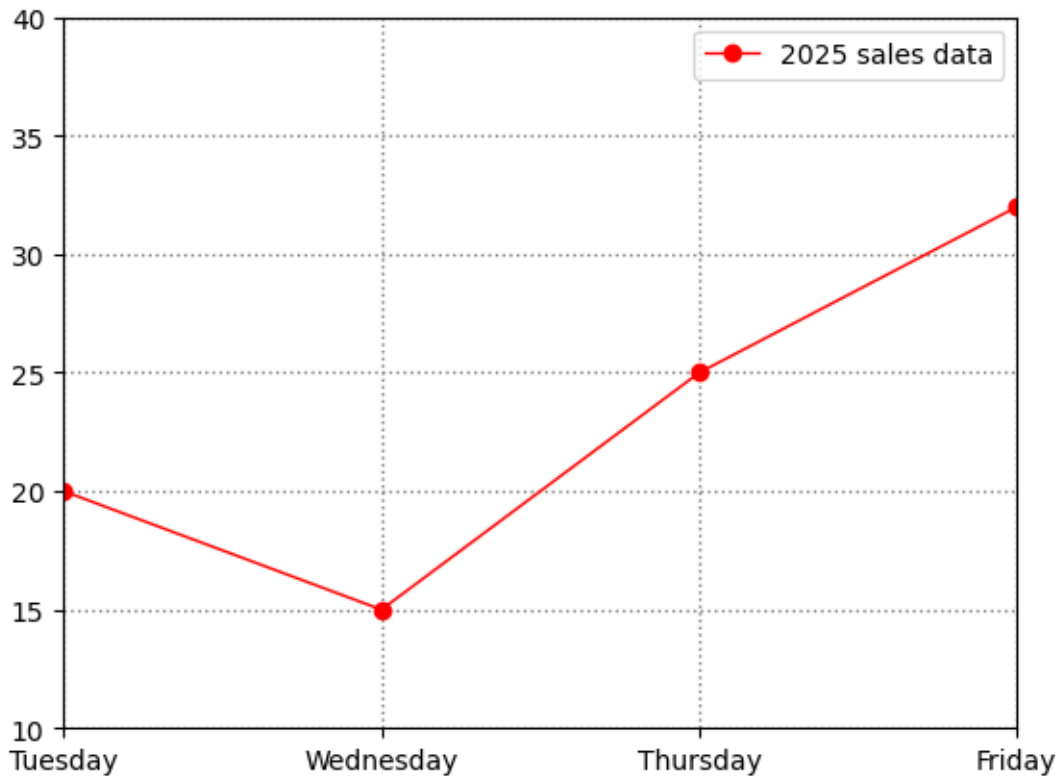


```
[12]: import matplotlib.pyplot as plt

# x=[1,2,3,4]
x=['Monday', 'Tuesday', 'Wednesday', 'Thursday', 'Friday', 'Saturday', 'Sunday']
y=[10,20,15,25,32,23,45]

plt.plot(x,y,color='red',linestyle='-',linewidth=1,marker='o',label='2025 sales_
data')

# plt.
    plot(x,y,color='red',linestyle='-',linewidth=1,marker='marker',label='label_name')
# plt.title("Bakery Products sell in a week!")
# plt.xlabel('Days')
# plt.ylabel('Cakes Sold')
plt.legend()
# plt.legend(loc='upper left/lower right',fontsize=12)
# plt.grid()
plt.grid(color='gray',linestyle=':',linewidth=1)
plt.xlim(1,4)
plt.ylim(10,40)
plt.show()
```



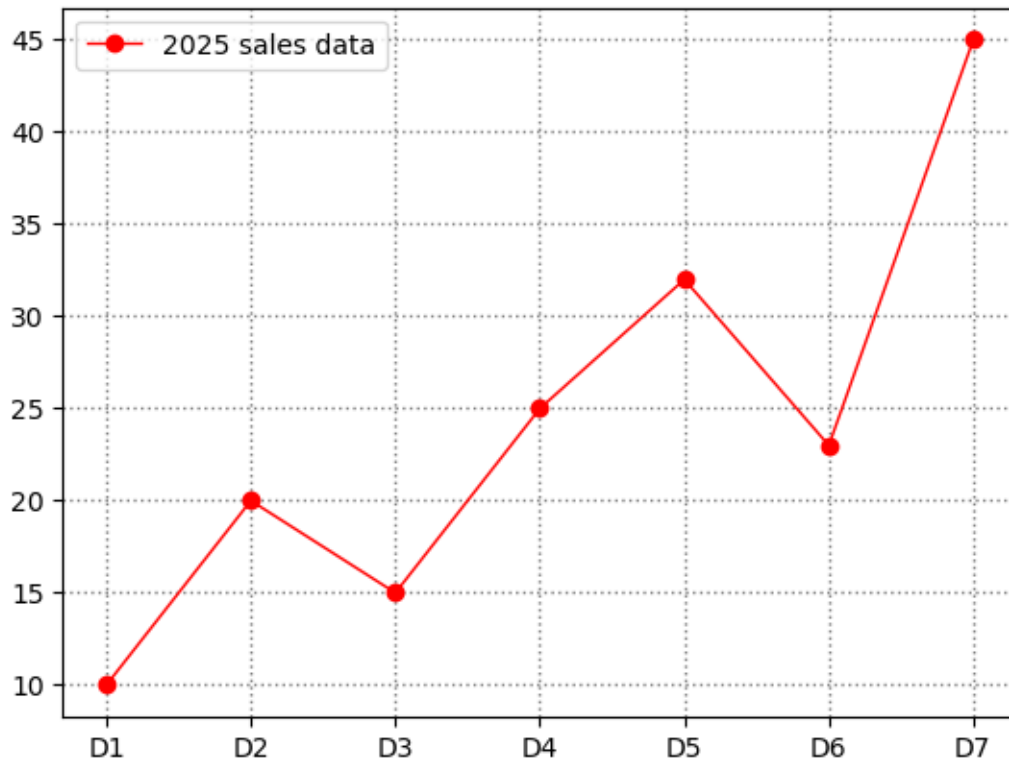
```
[14]: import matplotlib.pyplot as plt

# x=[1,2,3,4]
x=['Monday','Tuesday','Wednesday','Thursday','Friday','Saturday','Sunday']
y=[10,20,15,25,32,23,45]

plt.plot(x,y,color='red',linestyle='-',linewidth=1,marker='o',label='2025 sales_
↳data')

# plt.
↳plot(x,y,color='red',linestyle='-',linewidth=1,marker='marker',label='label_name')
# plt.title("Bakery Products sell in a week!")
# plt.xlabel('Days')
# plt.ylabel('Cakes Sold')
plt.legend()
# plt.legend(loc='upper left/lower right',fontsize=12)
# plt.grid()
plt.grid(color='gray',linestyle=':',linewidth=1)
# plt.xlim(1,4)
# plt.ylim(10,40)
plt.xticks([0,1,2,3,4,5,6],['D1','D2','D3','D4','D5','D6','D7'])
```

```
plt.show()
```

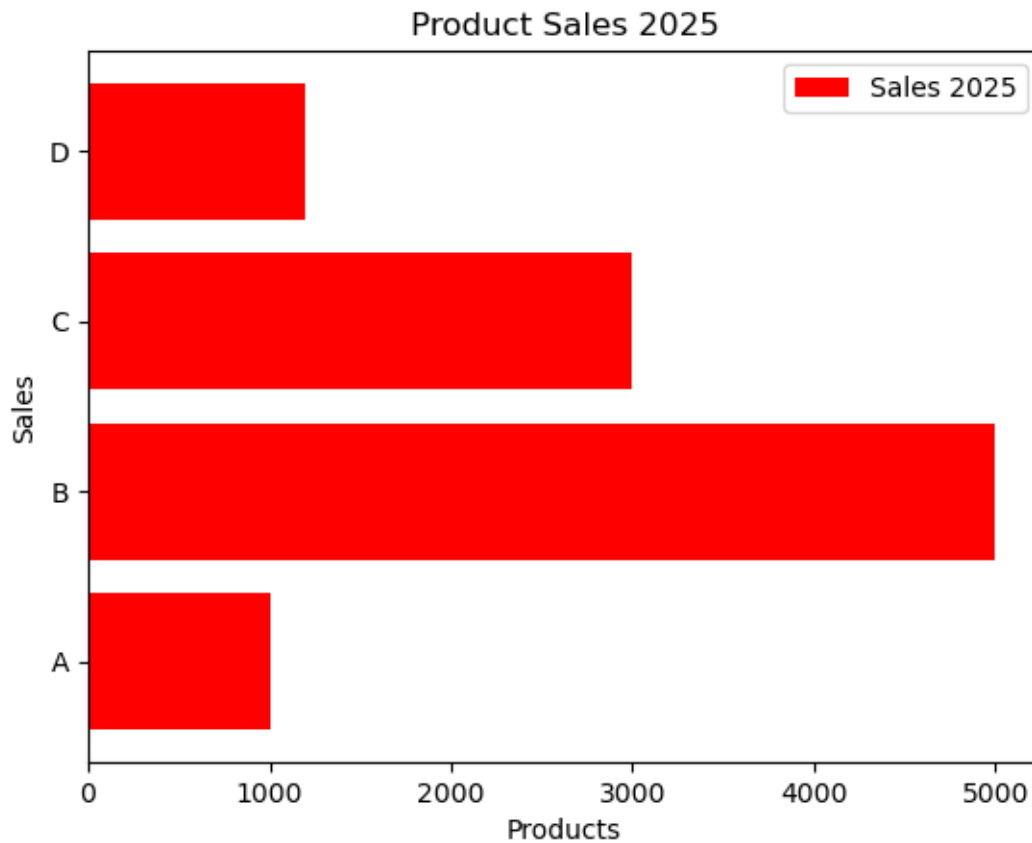


1

```
[2]: import matplotlib.pyplot as plt
product=['A','B','C','D']
sales=[1000,5000,3000,1200]
plt.bar(product,sales,color='red',label='Sales 2025')
plt.xlabel('Products')
plt.ylabel('Sales')
plt.title('Product Sales 2025')
plt.legend()
plt.show()
```



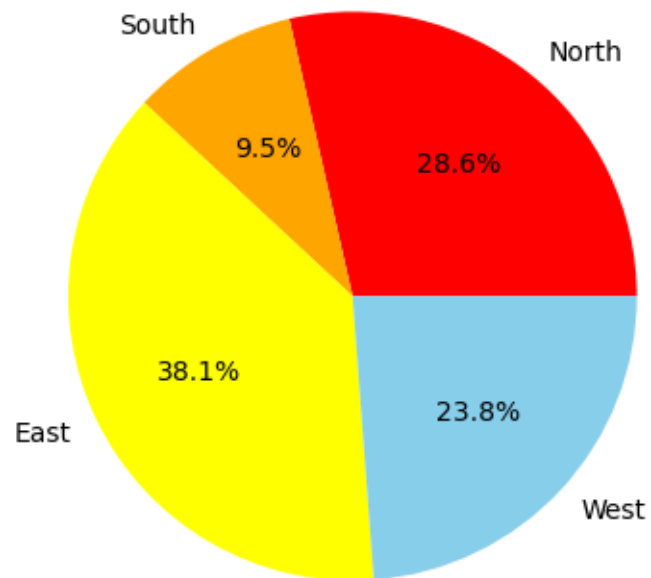

```
[3]: import matplotlib.pyplot as plt
product=['A','B','C','D']
sales=[1000,5000,3000,1200]
plt.barh(product,sales,color='red',label='Sales 2025')
plt.xlabel('Products')
plt.ylabel('Sales')
plt.title('Product Sales 2025')
plt.legend()
plt.show()
```



```
[6]: # plt.pie(value,label='list',color='list',autopct='%1.1f%%')
import matplotlib.pyplot as plt

regions=['North','South','East','West']
revenue=[3000,1000,4000,2500]
plt.pie(revenue,labels=regions,autopct='%1.
    1f%%',colors=['red','orange','yellow','skyblue'])
plt.title('REvenue contribution by Region')
plt.show()
```

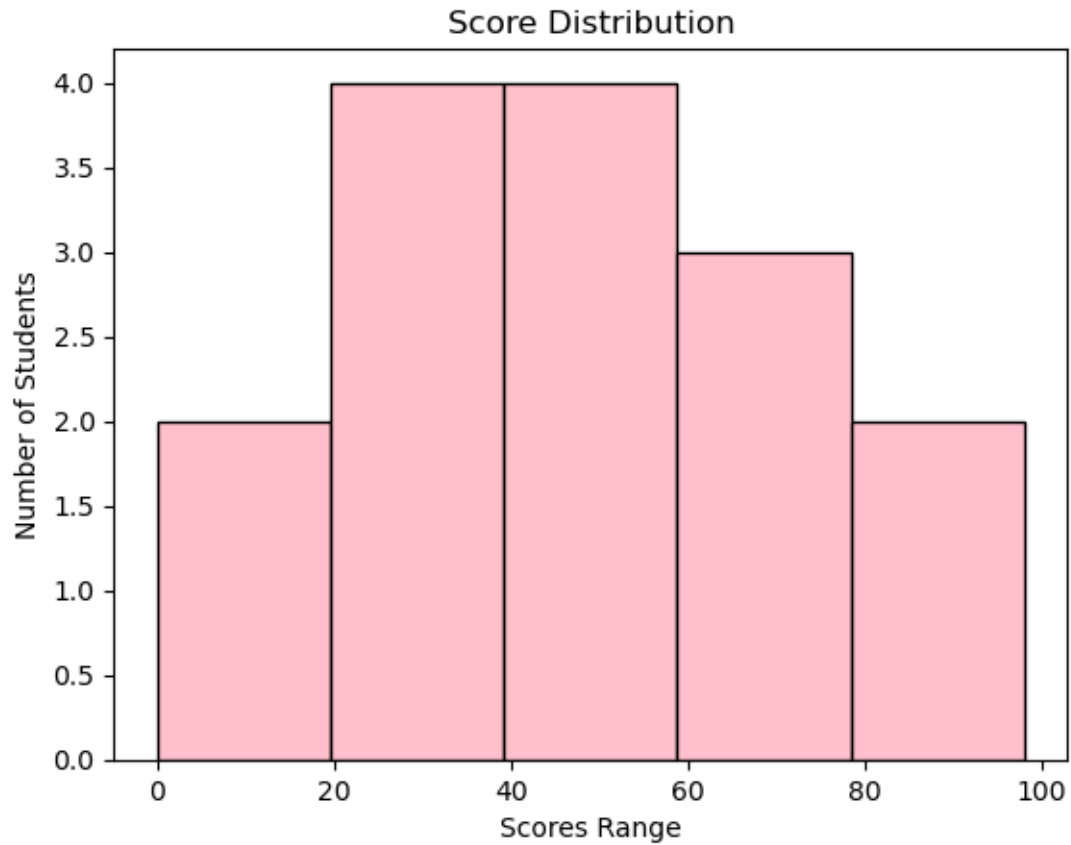
REvenue contribution by Region



```
[10]: #plt.hist(data,bins=number_of_bins,color='colorname',edgecolor='color')
import matplotlib.pyplot as plt

scores = [45, 67, 12, 34, 56, 78, 98, 54, 32, 51, 31, 21, 66,0,98]

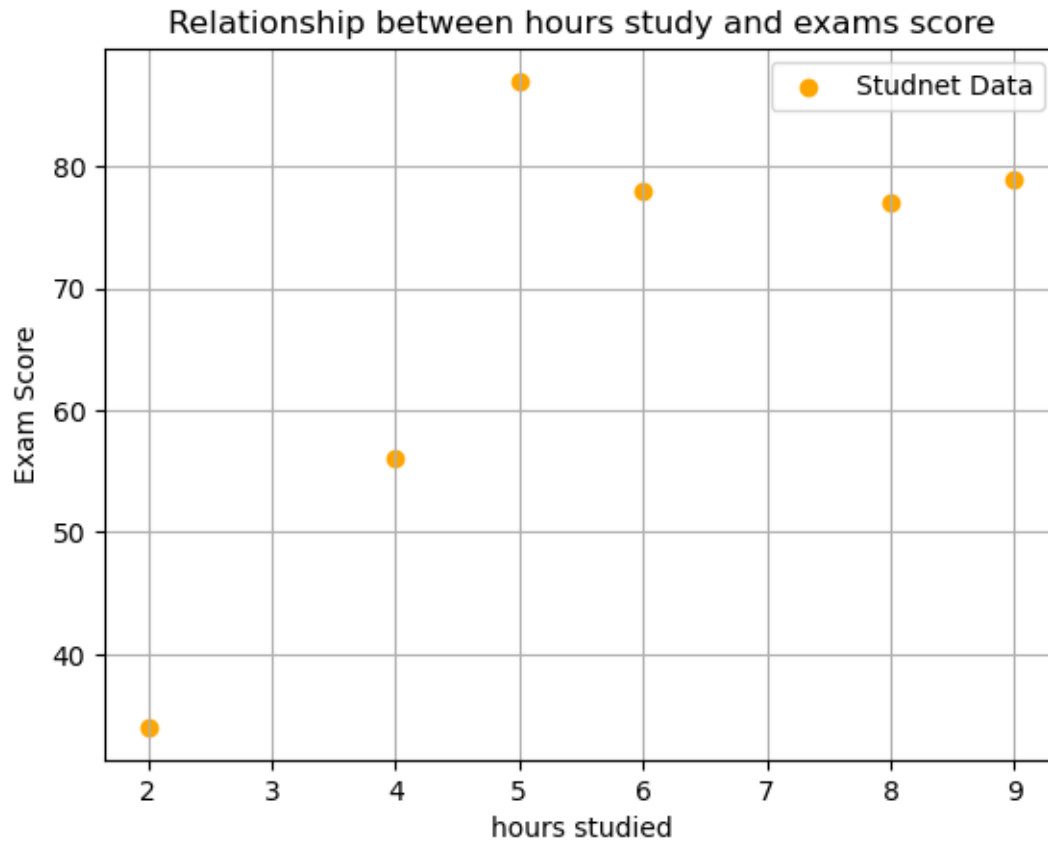
plt.hist(scores, bins=5, color='pink', edgecolor='black')
plt.xlabel('Scores Range')
plt.ylabel('Number of Students')
plt.title('Score Distribution')
plt.show()
```



```
[14]: #plt.scatter(x,y,color,marker,label)
import matplotlib.pyplot as plt

hours=[2,4,5,6,8,9]
score=[34,56,87,78,77,79]

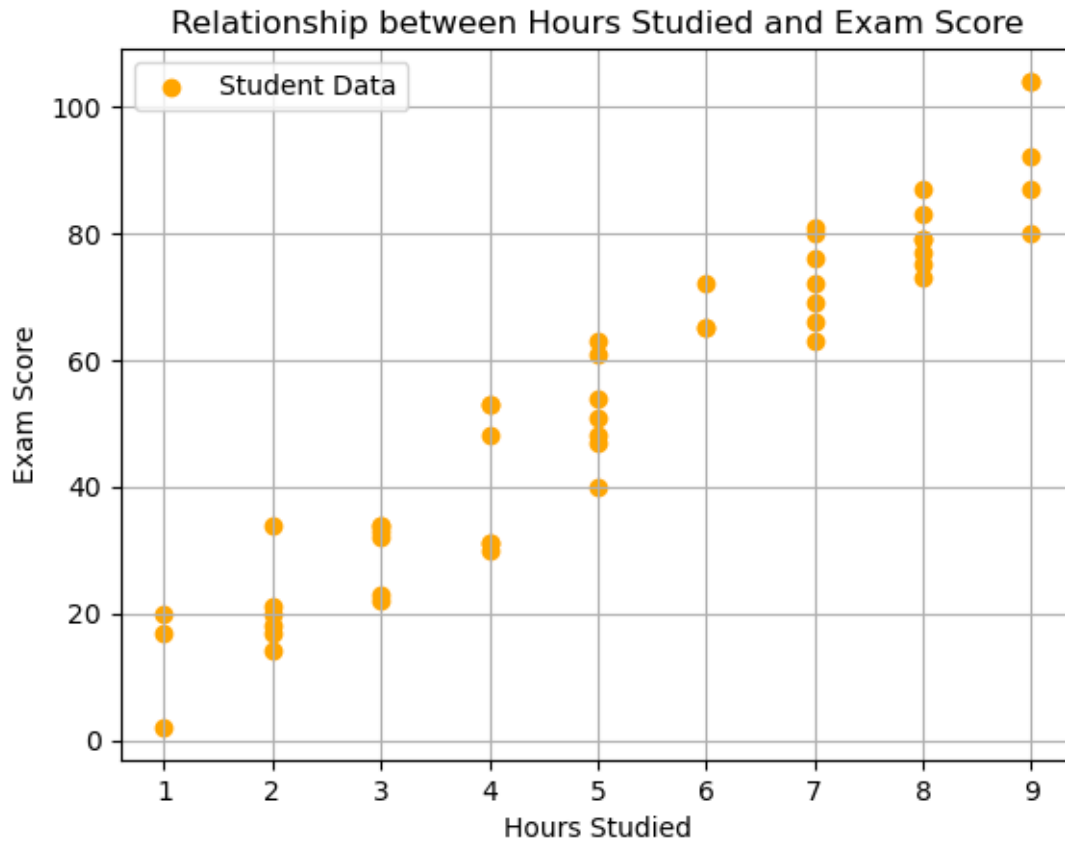
plt.scatter(hours,score,color='orange',marker='o',label='Studnet Data')
plt.xlabel('hours studied')
plt.ylabel('Exam Score')
plt.title('Relationship between hours study and exams score')
plt.legend()
plt.grid()
plt.show()
```



```
[13]: import matplotlib.pyplot as plt
import numpy as np

np.random.seed(42)
hours = np.random.randint(1, 10, 50)
score = hours * 10 + np.random.randint(-10, 15, 50)

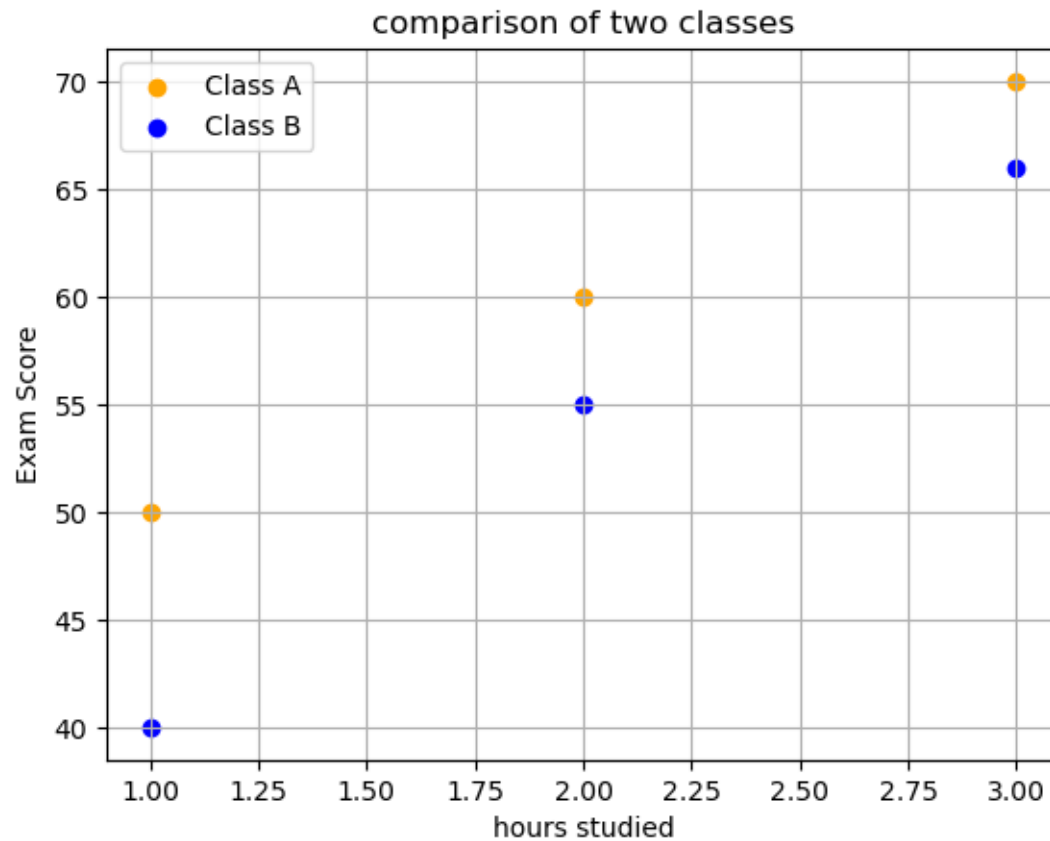
plt.scatter(hours, score, color='orange', marker='o', label='Student Data')
plt.xlabel('Hours Studied')
plt.ylabel('Exam Score')
plt.title('Relationship between Hours Studied and Exam Score')
plt.legend()
plt.grid()
plt.show()
```



```
[17]: import matplotlib.pyplot as plt

# hours=[2,4,5,6,8,9]
# score=[34,56,87,78,77,79]

plt.scatter([1,2,3],[50,60,70],color='orange',marker='o',label='Class A')
plt.scatter([1,2,3],[40,55,66],color='blue',marker='o',label='Class B')
plt.xlabel('hours studied')
plt.ylabel('Exam Score')
plt.title('comparison of two classes')
plt.legend()
plt.grid()
plt.show()
```



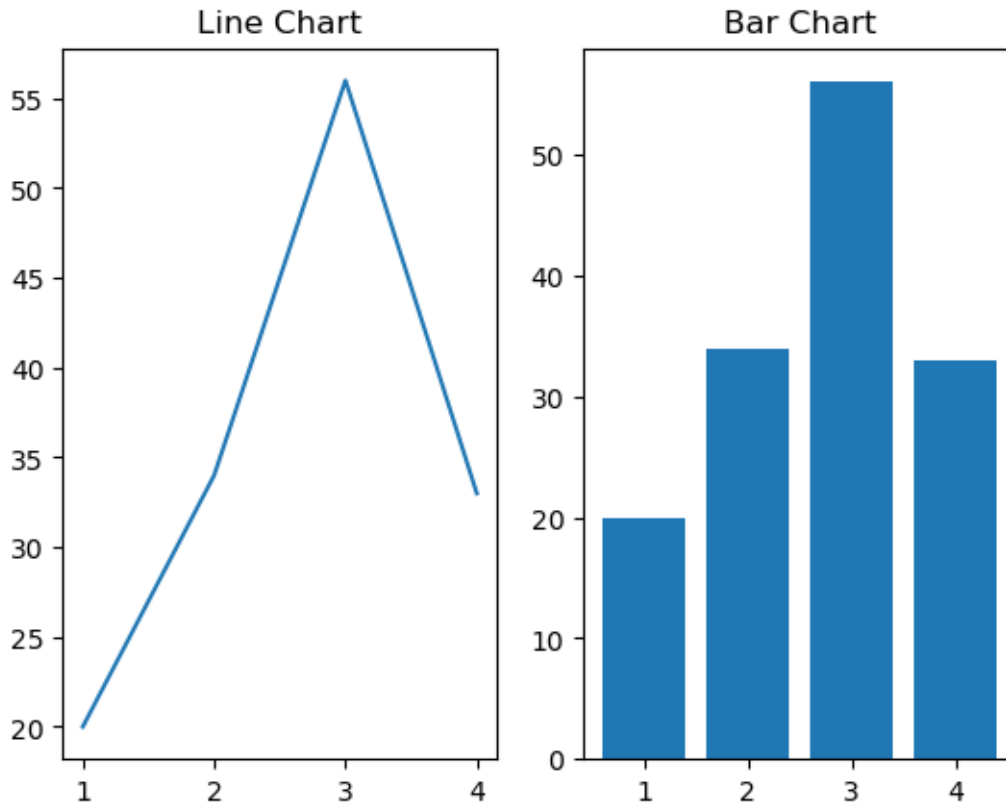
```
[19]: # plt.subplot(nrows,ncols,index)index start from 1
import matplotlib.pyplot as plt

x=[1,2,3,4]
y=[20,34,56,33]

plt.subplot(1,2,1)
plt.plot(x,y)
plt.title('Line Chart')

plt.subplot(1,2,2)
plt.bar(x,y)
plt.title('Bar Chart')

# plt.tight_layout()
plt.show()
```



```
[27]: # figure ax=plt.subplots(nrows,ncols,index,figsize=(width,height))index start_
      ↪from1
import matplotlib.pyplot as plt

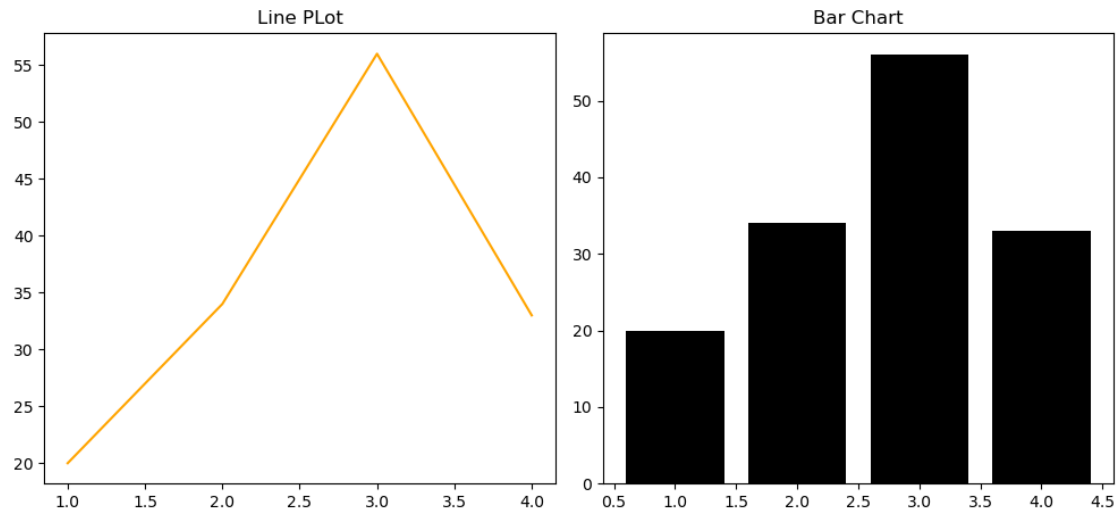
fig, ax=plt.subplots(1,2,figsize=(10,5))

x=[1,2,3,4]
y=[20,34,56,33]

ax[0].plot(x,y,color='orange')
ax[0].set_title('Line PLOT')

ax[1].bar(x,y,color='black')
ax[1].set_title('Bar Chart')
fig.suptitle('Comparison of Charts')
plt.tight_layout()
plt.show()
```


Comparison of Charts



[]: